

OPM Flow Reference Manual

2025-04

Open Porous Media

2025-01-01

Table of contents

1 OPM Flow Reference Manual

2025-04

1.1 About This Manual

The **OPM Flow Reference Manual** provides comprehensive documentation for the Open Porous Media (OPM) Flow reservoir simulator. OPM Flow is a fully-implicit, black-oil and compositional reservoir simulator capable of running industry-standard simulation models.

This manual covers:

- **Installation and Setup** – How to install and run OPM Flow
- **Keyword Reference** – Complete documentation of all supported keywords organized by input deck section (GLOBAL, RUNSPEC, GRID, EDIT, PROPS, REGIONS, SOLUTION, SUMMARY, SCHEDULE)
- **Appendices** – Keyword summaries, release notes, the OPMRUN graphical interface, Python scripting interface, command line options, and output file formats

1.1.1 How to Use This Manual

Use the table of contents or the search bar to navigate to specific keywords or topics. Keywords are organized by their input deck section, matching the structure of OPM Flow simulation input files.

Note

This manual corresponds to **OPM Flow version 2025-04**.

2 INTRODUCTION

2.1 Overview

The Open Porous Media (“OPM”) initiative was started in 2009 to encourage open innovation and reproducible research on modeling and simulation of porous media processes. OPM was initially founded as a collaboration between groups at Equinor (formerly Statoil), SINTEF, the University of Stuttgart, and the University of Bergen, but over time, several other groups and individuals have joined and contributed. What today forms the OPM suite of software, has mainly been developed by SINTEF, NORCE (formerly IRIS), Equinor, Ceetron Solutions (OPM Resinsight), Poware Software Solutions, Dr. Blatt HPC-Simulation-Software & Services, and OPM-OP. The initial vision was to create long-lasting, efficient, and well-maintained, open-source software for simulating flow and transport in porous media. The scope has later been extended to also provide open data sets, thus making it easier to benchmark, compare, and test different mathematical models, computational methods, and software implementations.

All software in OPM is distributed under the GNU General Public License (“GPL”), [version 3](#), with the option of using newer versions of the license. Datasets are distributed under the Open Data Commons Open Database License (“ODbL”), [version 1.0](#), with individual content under the Database Contents License (“DbCL”), [version 1.0](#).

The project’s numerical simulator, OPM Flow, is a fully-implicit, black-oil simulator capable of running industry-standard simulation models [Rasmussen, A.F., Sandve, T.H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkorn, R., Blatt, M., Rustad, A.B., Sævareid, O. et al. [2019]. The Open Porous Media Flow Reservoir Simulator. <https://doi.org/10.1016/j.camwa.2020.05.014>]. The simulator’s input file is compatible with the commercial simulator, although OPM Flow has extra keywords for supporting additional functionality not found in the commercial simulator. The simulator’s output files, except for the print and debug files, are compatible with the commercial simulator, and are readable by commercial post-processing software as well as by OPM ResInsight. OPM ResInsight is the Open Porous Media’s post-processing software.

The simulator is implemented using automatic differentiation to enable rapid development of new fluid models. In most simulators the coefficients of the linearized systems of equations (Jacobian matrix) required by the non-linear Newton Solver, have traditionally been computed by evaluating the closed-form expressions obtained by differentiating the discretized flow equations analytically. Differentiating these flow equations manually and programming the resulting formulas is time-consuming and error-prone. Thus, even simple extensions in functional dependencies can be difficult to accommodate into existing simulators. Automatic differentiation eliminates these drawbacks.

OPM Flow runs under Linux and Windows Subsystem for Linux, and the program is in active development with new features added in each bi-annual release. Currently the program has the following functionality:

2.1.0.1 Model Formulation

- Black-oil with dissolved gas and vaporized oil.
- Rock-dependent capillary and relative-permeability curves.
- End-point scaling and hysteresis.
- Oil vaporization controls (VAPPARS).
- Rock compaction.

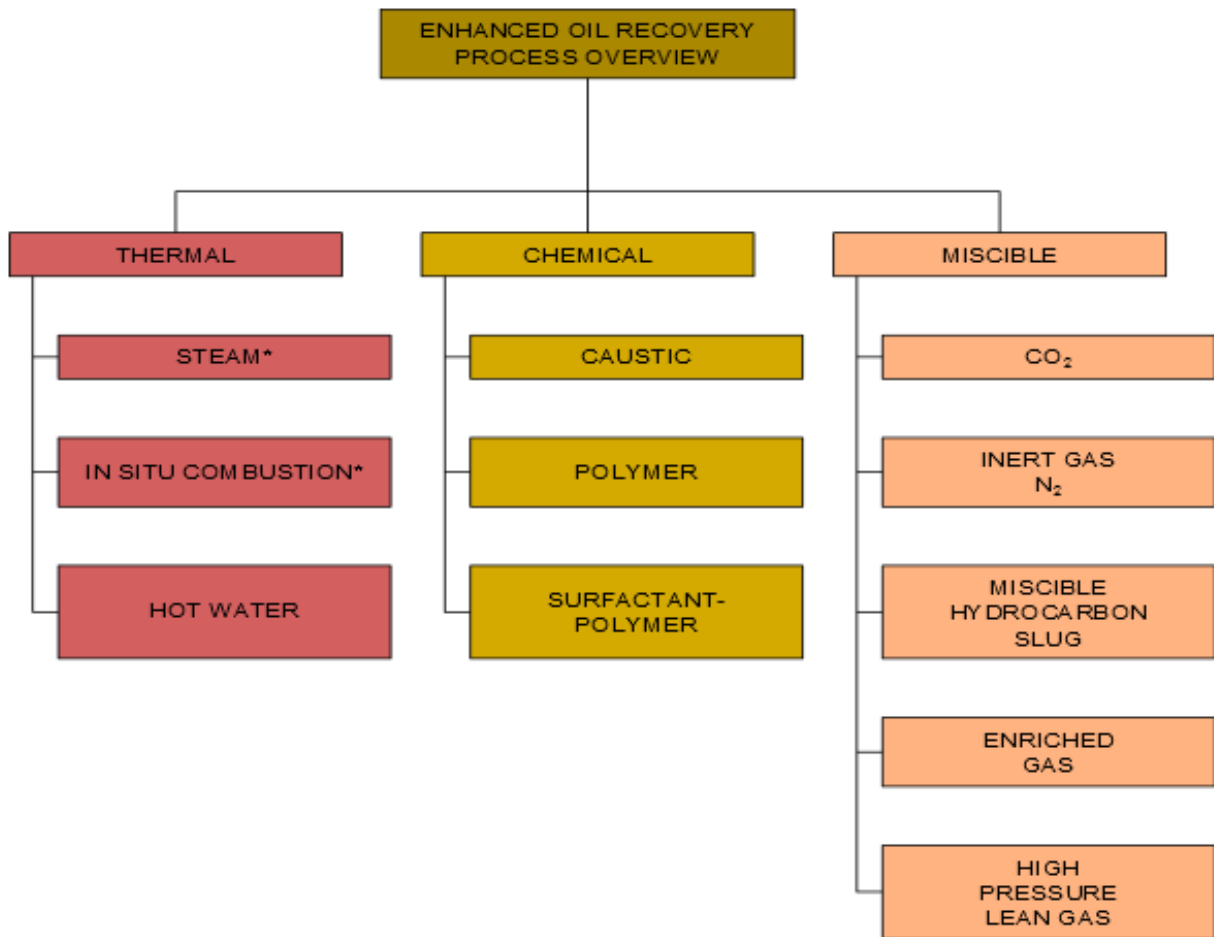
2.1.0.2 Enhance Oil Recovery Models

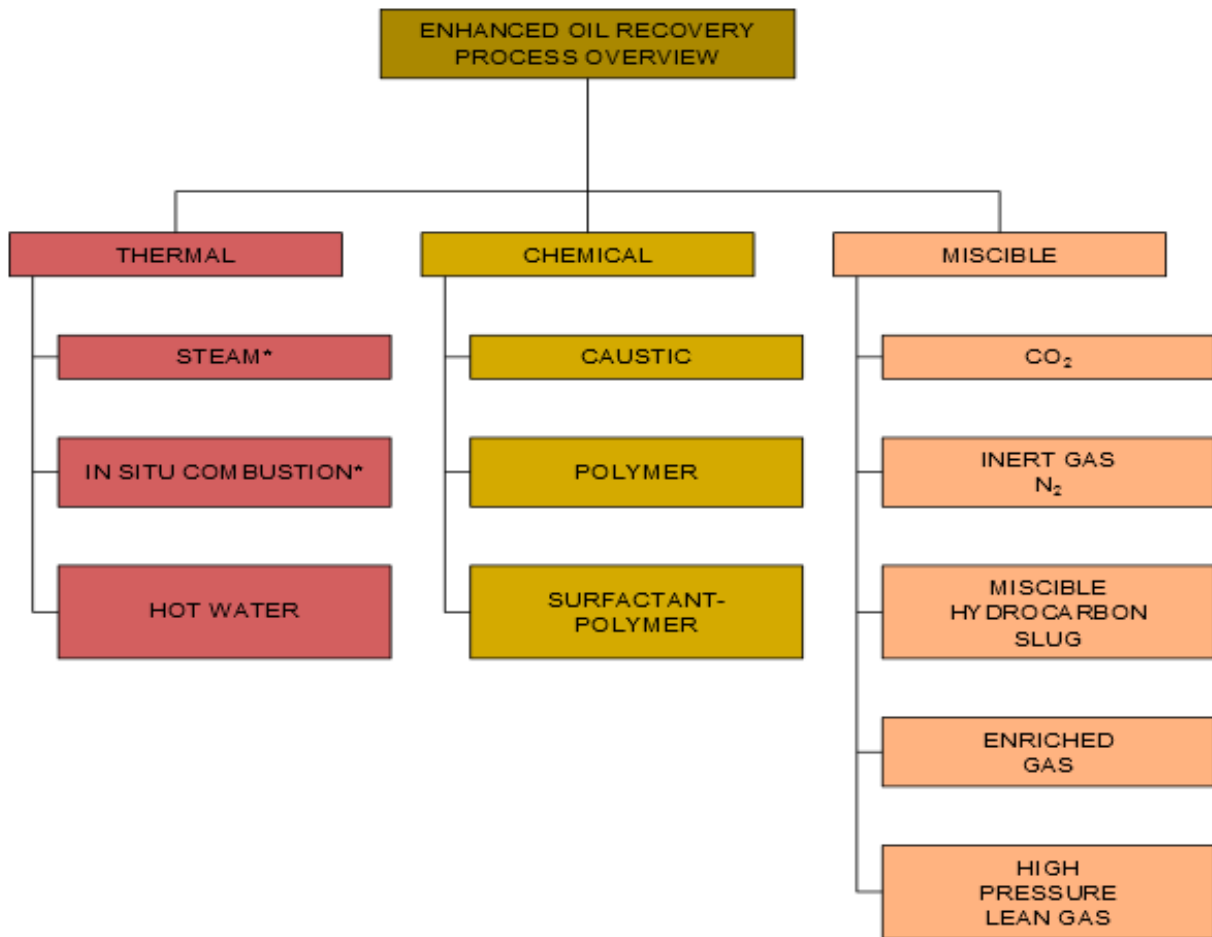
Enhanced Oil Recovery (“EOR”) is a term applied to methods used for recovering oil from a petroleum reservoir beyond that recoverable by primary and secondary methods. Primary recovery normally refers to production using the energy inherent in the reservoir from gas under pressure or a natural water drive. Secondary recovery usually refers to injection of water or water flooding. Thus, Enhanced Oil Recovery is often synonymous with tertiary recovery. Improved Oil Recovery (“IOR”) and Advanced Oil Recovery (“AOR”) have similar meaning, except they also apply to primary and secondary methods, and sometimes EOR methods can be used earlier in the sequence. Originally water flooding was considered as Enhanced Oil Recovery, but now EOR is generally considered to follow water flooding.

There are four major EOR groups of methods:

- thermal recovery,
- gas miscible recovery,
- chemical flooding, and
- microbial flooding.

These main groups are further subdivided into individual processes as illustrated in Figure 1.1. For example the thermal recovery methods consist of steam flooding and cyclic steam stimulation, as well as in-situ combustion. Figure 1.1 also includes hot water as a process for thermal EOR, but this process is rarely used in the industry as a drive mechanism, as steam contains significantly more heat than hot water and therefore steam is more effective in heating up the in situ oil. Note that microbial EOR methods, are not shown in Figure 1.1, due to their limited use in the industry.





Chemical flooding methods include polymer flooding (including polymer gels), micellar-polymer flooding, and alkaline flooding. Quite often these type of floods are combined to form a more efficient displacement process, for example in an Alkaline Surfactant Polymer (“ASP”) process.

Conventional gas flooding is an immiscible process, that is the oil and gas phases remain separate during the displacement drive. For the EOR gas drive process, the displacement is miscible, here the gas “mixes” with the oil to form a single phase fluid in the reservoir. Gas miscible processes are subdivided into carbon dioxide flooding, cyclic carbon dioxide stimulation, nitrogen flooding, and nitrogen-CO2 flooding.

OPM Flow has several EOR models based on extending the black-oil model with a fourth component in the simulator:

- **CO2 Standard EOR Model:** OPM Flow’s CO2 Standard EOR model is based on the simulators Solvent model by extending the black-oil formulation with a fourth component by adding a CO2 component to the gas phase. This is the standard formulation used by most black-oil simulators.
- **CO2 Dynamic EOR Model:** Although the CO2 Standard EOR model is commonly used, the formulation cannot account for the mass transfer of the various components and phases. The extended black-oil formulations often poorly represent the PVT properties of the oil-CO2 mixtures, resulting in poor agreement with the compositional formulation. OPM Flow’s

CO2 Dynamic EOR Model black-oil formulation attempts to overcome this limitation by making the black-oil properties dependent on the fraction of CO2 in the cell [T. H. Sandve, O. Sævareid and I. Aavatsmark: “Improved Extended Blackoil Formulation – for CO2 EOR Simulations.” in ECMOR XVII – The 17th European Conference on the – Mathematics of Oil Recovery, September 2020.]. This approach models the oil-CO2 mixture more accurately, and thus give results closer to the compositional simulator. Note the model is not restricted to CO2 utilization, as long as the appropriate oil-injection-gas dependent properties are entered in the model.

- **Foam Model:** An experimental foam module has been added to OPM Flow 2019-10 release. With this it is possible to simulate certain types of surfactant injection. Such injection stimulates formation of foam to change mobility ratios, and give better reservoir sweep. The implemented foam model treats surfactant transported in the gas or water phase, and reduces the mobility of that phase depending on the surfactant concentration. In addition to mobility reduction, foam adsorption to the reservoir rock is also included in the model.
- **Polymer Model (Standard):** OPM Flow’s polymer model is based on a black-oil polymer formulation, which is developed by extending the black-oil model with a polymer component. The effects of the polymer mixing are simulated based on the Todd-Longstaff [M. R. Todd and W. J Longstaff, The Development, Testing, and Application Of a Numerical Simulator for Predicting Miscible Flood Performance”. In: J. Petrol. Tech. 24.7 (1972), pages 874-882.] mixing model, and adsorption, dead pore space, and permeability reduction effects are also considered. A logarithmic shear thinning/thickening model has also been incorporated since the 2015-10 Release (see [Flow-polymer](#)). Note that the Polymer model has now been incorporated into the main OPM Flow simulator and is no longer a separate simulator.
- **Polymer Model (Molecular Weight Transport):** In addition to the standard polymer model, the simulator also has the Polymer Molecular Weight Transport option, that uses the polymer molecular weight in calculating the polymer viscosity. Here, the Mark-Howink equation, also known as the Mark-Howink-Staudinger equation [H. Mark, in R. Saenger, Der feste Koerper, Hirzel, Leipzig, 1938.], [R. Houwink , J. Prakt. Chem., Vol. 157, Issue 1-3, p. 15 (1940).], and [H. Staudinger, Die Hochmolekulare Organischen Verbindungen, Julius Springer, Berlin 1932.] is used to calculate the intrinsic viscosity as a function of the molecular weight of the polymer.
- **Solvent Model:** Similar to the polymer model, OPM Flow’s solvent model again uses an extra component by extending the black-oil formulation with a fourth component in the simulator by adding a solvent component to the gas phase (see [Flow-solvent](#)). Note that Solvent model has now been incorporated into the main OPM Flow simulator and is no longer a separate simulator.

2.1.0.3 CO2 Storage:

- **Brine-CO2 PVT** module that generates brine and CO2 PVT properties directly eliminating user input for PVT.
- Thermal properties generated by OPM Flow include the CO2 enthalpy represented as a table internally in the simulator. The liquid enthalpy depends on the dissolved CO2 and salinity as well as pressure and temperature. The water enthalpy is automatically calculated and modified to account for salinity and for CO2 following.

- Diffusion for fine grid modeling is implemented via coefficients that depend on temperature, pressure, and salinity. The diffusion coefficient is computed internally for pure water and modified to account for salinity. The effect of the porous media on the diffusion is modeled using an industry standard relationship. In addition, the coefficient can also be given as an input parameter using a keyword.
- Convective dissolution of CO₂ into in situ brine within a grid cell is implemented together with a convective dissolution parameter for controlling the rate of dissolution.
- Microbially Induced Calcite Precipitation (“MICP”) model for the modeling of biochemical processes in creating barriers by calcium carbonate cementation, the technology has the potential to be used for sealing leakage zones in geological formations [Landa-Marbán, D., Tveit, S., Kumar, K., Gasda, S.E., 2021. Practical approaches to study microbially induced calcite precipitation at the eld scale. *Int. J. Greenh. Gas Control* 106, 103256. [https://doi.org/10.1016/j.ijggc.2021.103256.](https://doi.org/10.1016/j.ijggc.2021.103256)] and [Landa-Marbán, D., Kumar, K., Tveit, S., Gasda, S.E., 2021. Numerical studies of CO₂ leakage remediation by micp-based plugging technology. In: Røkke, N.A. and Knuutila, H.K. (Eds) *Short Papers from the 11th International Trondheim CCS conference*, ISBN: 978-82-536-1714-5, 284-290.].

2.1.0.4 Other OPM Flow Specific Enhancements:

- The Salt Precipitation and Water Evaporation Model: The model accounts for water evaporation of the in situ brine and the resulting increasing dissolved salt concentration. Once the salt solubility limit is exceeded, salt precipitation occurs, that results in a decrease in porosity and permeability, and consequently a decrease in gas production. This feature is primary used in modeling gas fields under depletion.
- Thermal Model Enhancements: The energy black-oil implementation in OPM Flow is a mixture of the commercial simulators black-oil temperature and compositional thermal options. These commercial simulator options are two separate modeling facilities in the commercial simulator. OPM Flow’s thermal implementation is based on solving the energy equation fully coupled with the black-oil equations so the results are not directly equivalent to the commercial simulator’s black-oil temperature or compositional thermal formulations. In addition, OPM Flow can take into account the Joule–Thomson effect for oil, gas, and water. This behavior is especially important in gas wells, when a real gas expands, resulting in the temperature of the gas dropping.
- Biofilm Effects in Underground Applications: Biofilm-related effects in subsurface applications such as hydrogen storage include reduced injectivity and hydrogen loss. The conceptual model includes the following mechanisms: (1) Biofilm is present in the storage site prior to injection and (2) the biofilm consumes injected H₂/CO₂, leading to clogging effects.

2.1.0.5 Description of Geology:

- Rectilinear and fully-unstructured grid.
- Corner-point grids from the commercial simulator input, including fault and region multipliers, minimum pore volume, pinch outs, etc.
- Spider grids that are similar to radial grids for modeling near well bore effects.
- Analytical and numerical aquifers.

2.1.0.6 Well and Group Controls:

- Bottom-hole pressure and surface/reservoir rate.
- Group controls, including voidage replacement, network model, gas lift optimization, guide rate control for producers and injection groups and well, etc.
- Shut/stop/open individual completions.
- History-matching wells.
- Multi-segment wells together with various inflow control devices (“ICDs”).
- Well lists.
- Action conditions and command processing via the ACTIONX and UDQ keywords employed in the commercial simulator.
- Python interface to control and process simulator data.

2.1.0.7 Input and Output:

- General reader/parser for Eclipse input decks.
- XML-based or simple text-format input of additional parameters.
- Flexible output of summary and restart files in the commercial simulator’s format.
- Logging to terminal and print file.
- Front-end graphical interface for running the simulator with various tools, including: case compression and uncompression, keyword generation etc.

2.1.0.8 Simulation Technology:

- Fully-implicit in time.
- Automatic differentiation.
- Two-point flux approximation in space with upstream-mobility weighting.
- Flexible assembly through the use of automatic differentiation.
- Bi-conjugate Gradient method (BiCG-stab) and a restarted Generalized Minimal Residual (“GMRes”) iterative linear solvers with ILU0 and Constraint Pressure Residual (“CPR”) preconditioners.
- Adaptive step-size controls.
- Parallel processing.

The following section briefly describe the simulator’s formulation, equation, and the various numerical procedures available to solve the equations.

Note

The material in the following sections is taken from Rasmussen et al. [Rasmussen, A.F., Sandve, T.H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkorn, R., Blatt, M., Rustad, A.B., Sævareid, O. et al. [2019]. The Open Porous Media Flow Reservoir Simulator. <https://doi.org/10.1016/j.camwa.2020.05.014>.], and the reader is encouraged to read the full

paper to obtain a fuller understanding of the simulator's formulation and the resulting equations.

2.2 Black-Oil Model Equations

The black-oil equations constitute the most widely used flow model in the simulation of hydrocarbon reservoirs. The model is based on the premise that we have three different fluid phases (aqueous, oleic, and gaseous), and three (pseudo) components: water, oil, and gas. Oil and gas are not single hydrocarbon species, but represent all hydrocarbon species that exist in liquid and vapor form at surface conditions. Mixing is allowed, in the sense that both oil and gas can be found in the oleic phase, in the gaseous phase, or in both. The amount of dissolved gas in the oleic phase and vaporized oil in the gaseous phase must be kept track of on a grid cell basis. In the following discussions the subscripts w represents the water, o the oil, and g the gas quantities related to each of the phases or components.

The black-oil equations can be deduced from conservation of mass for each component with suitable closure relationships such as Darcy's law and initial and boundary conditions. The equations are discretized in space with an upstream finite-volume scheme using a two-point flux approximation, and in time using an implicit (backward) Euler scheme. The resulting equations are solved simultaneously in a fully implicit formulation by a Newton-type linearization with a properly preconditioned, iterative linear solver (see Rasmussen et al. [Rasmussen, A.F., Sandve, T.H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkorn, R., Blatt, M., Rustad, A.B., Sævareid, O. et al. [2019]. The Open Porous Media Flow Reservoir Simulator. <https://doi.org/10.1016/j.camwa.2020.05.014>.]).

The conservation laws form a system of partial differential equations, one for each pseudo component :

$$\frac{\partial}{\partial t} (\phi_{ref} A_\alpha) + \nabla \cdot \mathbf{u} + q_\alpha = 0 \quad (2.1)$$

where the accumulation terms and fluxes are given by:

$$A_w = m_\phi b_w s_w \quad (2.2)$$

$$A_o = m_\phi (b_o s_o + r_{og} b_g s_g) \quad (2.3)$$

$$A_g = m_\phi (b_g s_g + r_{go} b_o s_o) \quad (2.4)$$

and the phase fluxes are given by Darcy's law:

$$\mathbf{v} = -\lambda_\alpha \mathbf{K} (\nabla p_\alpha - \rho_\alpha \mathbf{g}) \quad (2.5)$$

In addition, the following closure relations should hold:

$$s_w + s_o + s_g = 1 \quad (2.6)$$

$$p_{c,ow} = p_o - p_w \quad (2.7)$$

$$p_{c,og} = p_o - p_g \quad (2.8)$$

where:

ϕ = porosity, can be a function of (oil) pressure: $\phi = m_\phi(p_o) \phi_{ref}$.

m_ϕ = pore volume multiplier as function of pressure.

ϕ_{ref} = reference porosity, a constant in time but varying in space.

p_α = pressure for phase α .

b_α = shrinkage/expansion factor for phase α defined as the ratio of the surface

volume at standard conditions to the reservoir volume for a given amount of fluid: $b_\alpha = V_{surface;\alpha}/V_{reservoir;\alpha}$. For the oil phase, b_o is called the shrinkage factor, whereas for gas b_g is called the expansion factor. The reciprocal quantity is called the formation volume factor, and is usually denoted with a capital B: $B_\alpha = 1/b_\alpha$. The variable is usually a function of phase pressure and composition: $b_\alpha = b_\alpha(p_\alpha, r_{go}, r_{og})$.

r_{go} = gas-oil ratio ("GOR"), ratio of dissolved gas to oil in the oleic phase,

commonly called r_s or R_s in other literature.

r_{og} = oil-gas ratio or condensate-gas ratio ("CGR"), ratio of vaporized oil to gas

in the gaseous phase, commonly called r_v or R_v in other literature.

s_α = saturation of phase α , the pore volume fraction occupied by the phase.

The saturations of all phases sum to one.

$p_{c,\alpha\beta}$ = capillary pressure between phases α and β . Typically a function of saturation: $p_{c,\alpha\beta} = p_{c,\alpha\beta}(S_\alpha)$.

K = permeability of the porous medium.

$k_{r,\alpha}$ = relative permeability for phase α modeling the reduction in effective permeability for a fluid phase in the presence of other phases. Typically a function of saturation.

μ_α = viscosity of phase α , typically a function of phase pressure and composition.

λ_α = mobility of phase α , given by $\lambda_\alpha = k_{r,\alpha}/\mu_\alpha$.

v = velocity of phase α .

u = velocity of component α .

$\rho_{S,\alpha}$ = surface density of phase α at one atmosphere, a given constant.

ρ_α = density of phase α in the reservoir. For water, $\rho_w = b_w \rho_{S,w}$. For the oleic

phase the relationship is more complex since it must include dissolved gas: $\rho_o = b_o (\rho_{S,o} + r_{go} \rho_{S,g})$, and similar for the gaseous phase we have: $\rho_g = b_g (\rho_{S,g} + r_{og} \rho_{S,o})$.

g = gravitational acceleration vector.

q_α = well out flux density of pseudo component α , (negative for well in-flow).

The form of this term depends on the well model used, either the standard well model or the multi-segment model.

Equations (1.1) to (1.8) are then transformed to the discrete form using a forward finite difference scheme. In the following, let subscript i denote a discrete quantity defined in cell i and subscript ij denote a discrete quantity defined at the connection between two cells i and j . Two cells share a connection if they are adjacent geometrically in the computational grid or share an explicit Non-Neighbor Connection ("NNC") [Connections between cells due to geometric adjacency are automatically calculated by the simulator, including across faults in the grid. In addition, the user can specify extra connections between arbitrary cells using the NNC keyword in the GRID section.]. For example $v_{o,ij}$ is the oil flux from cell i to cell j . For oriented quantities such as fluxes, the orientation is taken to be from cell i to cell j , and the quantity is skew-symmetric ($v_{o,ij} = -v_{o,ji}$), whereas non-oriented quantities such as the transmissibility T_{ij} are symmetric ($T_{ij} = T_{ji}$). Quantities with superscript 0 are taken at the start of the discrete time step, other quantities are at the end of the time step. Superscripts or subscripts applied to an expression in parenthesis apply to each element in the expression.

The discretized equations and residuals are, for each pseudo-component α and cell i :

$$R_{\alpha,i} = \frac{\phi_{ref,i} V_i}{\Delta t} (A_{\alpha,i} - A_{\alpha,i}^0) + \sum_{j \in C(i)} u_{\alpha,ij} + q_{\alpha,i} = 0 \quad (2.9)$$

where the accumulation terms are the same as equations (1.2) to (1.4) and fluxes u_α are of similar form to the velocities, u in the aforementioned equations; thus we have

$$A_{wi} = (m_\phi b_w s_w)_i \quad (2.10)$$

$$A_{oi} = (m_\phi (b_o s_o + r_{og} b_g s_g))_i \quad (2.11)$$

$$A_{gi} = (m_\phi (b_g s_g + r_{go} b_o s_o))_i \quad (2.12)$$

In addition, equations (1.6) to (1.8) also hold for each cell i , that is:

$$(s_w + s_o + s_g)_i = 1 \quad (2.13)$$

$$(p_{c,ow})_i = (p_o - p_w)_i \quad (2.14)$$